Supporting Information: Supporting Information: Target breadth and mutation resilience in African-natural-product-inspired antimalarial chemotypes: a computational analysis

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This supplement presents the numerical material for the integrated computational analysis, organized by evidence layer: chemical space, target-anchored docking, RRS, and exploratory cross-metric analysis. The four-target matrix is reported target by target; no cross-target mean is used because the Vina score is not calibrated across unlike receptor pockets.

S1 Cohort and evidence boundaries

The integrated analysis uses the 17-member Set-C polypharmacology-oriented cohort described in the main text (Section 2.2). Set C was selected by weighted MPO, synthetic accessibility, QED, and target-profile criteria; all 17 entries carried $n_{\text{targets bound}} = 2$ in the source selection table. Set C is therefore an enriched cohort, which limits prevalence-based interpretation: the $n_{\text{targets bound}} = 2$ label is a polypharmacology selection criterion, not a constraint on the 17-by-4 wild-type docking profile reported here. Docking follows AutoDock Vina conventions.[S1,S2]

The MPO top-20 and the MD top-20 are distinct sets, and the MD validation systems 201, 438, the 164-labelled PfClpP system, and 214 are not MD validation of Set C. RRS comes from the separate WT/mutant docking panel and is available only for the PfDHFR and PfCRT mutant panels.

S2 Chemical-space coverage and target-anchored docking

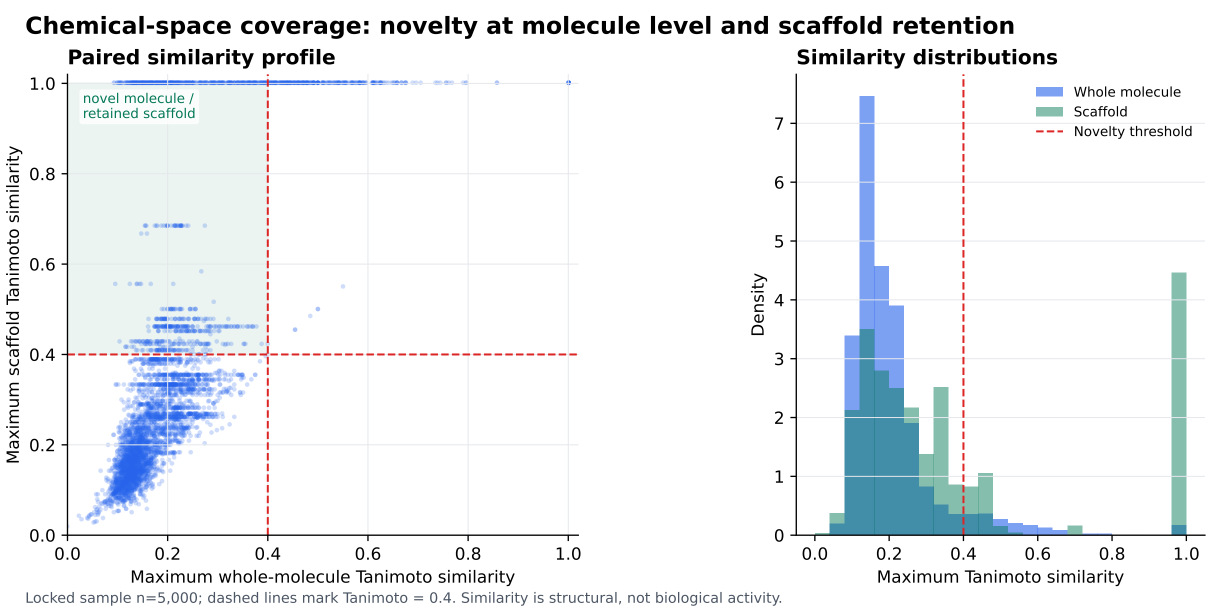

Figure S1: Chemical-space coverage in the 5 000-molecule novelty-analysis sample. The paired plot compares maximum whole-molecule and scaffold Morgan-fingerprint Tanimoto similarities to the African-natural-product reference set; the dashed lines mark the operational threshold of 0.4. The upper-left region represents molecules that are dissimilar at the whole-molecule level while retaining scaffold-level similarity. Similarity is a structural descriptor and does not indicate biological activity, target preference, or synthetic success.

S3 Target-anchored docking matrix

Table S1: Target-anchored AutoDock Vina score estimates for the enriched 17-member Set-C cohort. Values are in kcal mol^{-1}; they are scoring-function outputs, not experimental free energies. Columns are target by target (no cross-target averaging). The PfCRT column lists the re-docked values against the corrected 3D7-like LYS-76 receptor (Section S12.7), used in the main-text RRS and favorability analyses (Table 1; Table 2).

Candidate	PfDHFR	PfCRT	PfClpP	PfATP4
PP-01	−5.947	−9.155	−5.467	−6.361
PP-02	−5.562	−9.337	−5.907	−6.591
PP-03	−6.089	−12.010	−6.358	−7.311
PP-04	−5.493	−7.398	−5.386	−6.086
PP-05	−6.285	−8.959	−6.466	−6.378
PP-06	−6.267	−9.027	−7.031	−7.285
PP-07	−6.226	−7.952	−6.226	−5.775
PP-08	−4.935	−6.592	−5.052	−5.494
PP-09	−5.693	−7.436	−5.613	−6.078
PP-10	−6.346	−8.039	−6.016	−6.670
PP-11	−6.179	−9.343	−6.583	−6.724
PP-12	−5.248	−8.056	−5.386	−4.635
PP-13	−5.921	−9.788	−6.454	−7.565
PP-14	−4.863	−6.431	−5.047	−5.460
PP-15	−6.045	−9.808	−6.366	−7.016
PP-16	−5.469	−7.469	−6.203	−6.451
PP-17	−5.266	−6.374	−5.342	−5.360

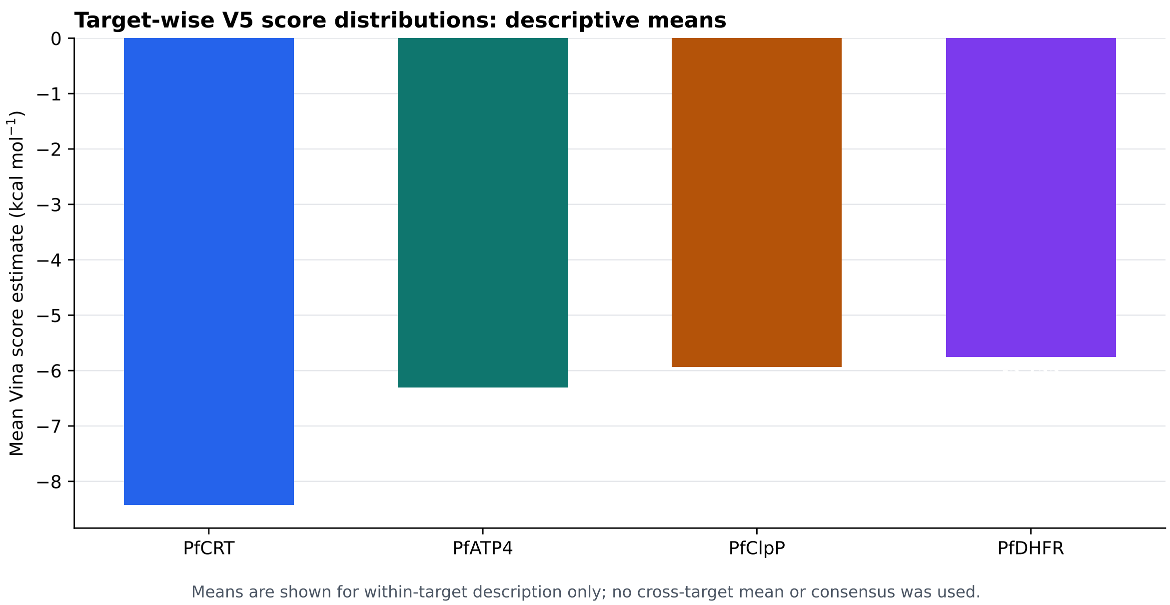

Figure S2: Descriptive target-wise means from the raw docking panel. Bars show the within-target mean across the 17 candidates. Means are shown only within target; no cross-target arithmetic mean or consensus score was used. These summaries are computational scoring-function outputs, not experimental affinity measurements.

S4 Physicochemical properties of Set-C candidates

Table S2: Physicochemical properties of Set-C polypharmacology candidates (PP-01 to PP-17). MW: molecular weight (Da); logP: octanol-water partition coefficient; HBD: hydrogen bond donors; HBA: hydrogen bond acceptors; TPSA: topological polar surface area (Å^2); RB: rotatable bonds; QED: quantitative estimate of drug-likeness (0-1 scale); SA: synthetic accessibility (1 = easy to synthesize, 10 = difficult); Lipinski: number of Lipinski Rule of Five violations.

Candidate	MW	logP	HBD	HBA	TPSA	RB	fsp^3	QED	SA	Lipinski
PP-01	328.3	3.19	1	6	78.1	4	0.17	0.79	2.2	0
PP-02	388.5	3.09	1	6	66.4	8	0.45	0.75	3.3	0
PP-03	356.4	3.38	4	6	107.2	3	0.25	0.63	3.4	0
PP-04	259.3	3.01	0	5	53.7	3	0.21	0.72	2.5	0
PP-05	284.3	2.88	2	5	79.9	2	0.06	0.76	2.1	0
PP-06	354.4	4.32	2	5	76.0	4	0.29	0.80	3.1	0
PP-07	230.3	3.14	0	3	39.4	3	0.21	0.60	2.2	0
PP-08	222.2	2.16	0	4	36.9	4	0.33	0.73	2.6	0
PP-09	185.3	3.68	1	0	15.8	2	0.23	0.69	2.4	0
PP-10	242.2	2.66	1	4	59.7	1	0.07	0.67	2.0	0
PP-11	330.3	2.59	3	7	109.4	3	0.12	0.68	2.4	0
PP-12	222.4	4.40	1	1	20.2	7	0.60	0.63	3.5	0
PP-13	330.3	2.05	3	7	113.3	2	0.18	0.72	3.3	0
PP-14	180.2	2.14	2	3	49.7	2	0.20	0.69	2.3	0
PP-15	316.3	2.03	3	6	96.2	3	0.24	0.80	3.4	0
PP-16	228.2	2.07	1	4	67.5	0	0.08	0.64	2.6	0
PP-17	164.2	1.49	2	2	57.5	2	0.00	0.65	1.8	0
Mean	271.9	2.84	1.6	4.4	66.3	3.1	0.22	0.703	2.7	0.0

S5 Drug-likeness compliance of Set-C candidates

S6 ADMET profile summary

S7 RRS definition and class summary

For candidate i, mutation m, and target t (Equation (1); Section 2.4):

$$\mathrm{RRS}_{i,m,t} = 100|S_{\mathrm{Vina},i,m,t}|/|S_{\mathrm{Vina},i,WT,t}|.$$

Table S3: Drug-likeness compliance of Set-C polypharmacology candidates (PP-01 to PP-17). Lipinski: Lipinski Rule of Five (MW $\leq$ 500, logP $\leq$ 5, HBD $\leq$ 5, HBA $\leq$ 10); Veber: Veber rules (RB $\leq$ 10, TPSA $\leq$140 Å^2); QED: quantitative estimate of drug-likeness (High $>$ 0.70, Moderate $\leq$ 0.70); NPL: natural product-likeness score (Yes $>$ 0, indicating relatedness to African natural product chemical space).

Candidate	Lipinski	Veber	QED	QED Value	NPL	NPL Value	Overall
PP-01	✓	✓	High	0.79	Yes	0.67	Full
PP-02	✓	✓	High	0.75	Yes	0.61	Full
PP-03	✓	✓	Moderate	0.63	Yes	0.62	Partial
PP-04	✓	✓	High	0.72	Yes	0.68	Full
PP-05	✓	✓	High	0.76	Yes	0.76	Full
PP-06	✓	✓	High	0.80	Yes	0.62	Full
PP-07	✓	✓	Moderate	0.60	Yes	0.59	Partial
PP-08	✓	✓	High	0.73	Yes	0.56	Full
PP-09	✓	✓	Moderate	0.69	Yes	0.64	Partial
PP-10	✓	✓	Moderate	0.67	Yes	0.78	Partial
PP-11	✓	✓	Moderate	0.68	Yes	0.67	Partial
PP-12	✓	✓	Moderate	0.63	No	0.00	Partial
PP-13	✓	✓	High	0.72	Yes	0.67	Full
PP-14	✓	✓	Moderate	0.69	Yes	0.46	Partial
PP-15	✓	✓	High	0.80	Yes	0.70	Full
PP-16	✓	✓	Moderate	0.64	Yes	0.76	Partial
PP-17	✓	✓	Moderate	0.65	Yes	0.50	Partial
Compliant	17/17	17/17	8/17	—	16/17	—	8/17
Percentage	100 %	100 %	47 %	—	94 %	—	47 %

Here S_{Vina} denotes an AutoDock Vina scoring-function output, not an experimental free energy. RRS eligibility is determined from the separate WT/mutant docking panel, not from the four-target wild-type matrix. Only wild-type binders with $|S_{\text{Vina},i,WT,t}| \geq 5\,\text{kcal}\,\text{mol}^{-1}$ were eligible. The 17-candidate class counts were A*: 7, B: 4, C: 6, and D: 0. The observed RRS range was 73.2 to 115.6. The wild-type and mutant docking scores are in Table S6; machine-readable values are in the Zenodo deposit.

S8 Exploratory cross-metric analysis

PNS, ACSI, and favorability follow the main-text definitions (Section 2.5). The N_{fav}–RRS association was further assessed by a permutation test (100 000 random RRS assignments):

Table S4: ADMET profile summary for Set-C polypharmacology candidates. Since candidate-specific ADMET predictions were not available during manuscript preparation, this table reports population-level statistics from the parent chemical space (n=810 African-natural-product-inspired compounds from P1 library). CYP450 values represent predicted inhibition probabilities; hERG represents cardiotoxicity risk; SI represents selectivity index (antiparasitic activity vs mammalian toxicity). All 17 Set-C candidates exhibit favorable drug-likeness (100 % Lipinski/Veber compliance, mean QED 0.703), suggesting low-to-moderate ADMET risk. Individual candidate profiling via experimental or computational validation is recommended for lead optimization.

ADMET Property	Population Mean	Interpretation
CYP450 Inhibition Probability		
CYP2C9	26.1 %	Low-moderate risk
CYP2C19	42.4 %	Moderate risk
CYP3A4	42.6 %	Moderate risk
CYP2D6	17.0 %	Low risk
Cardiotoxicity & Safety		
hERG inhibition	11.7 % (mean)	Low risk
hERG inhibition	11.8 % (median)	Low risk
Selectivity		
Antiparasitic SI > 10	100 % ($n = 810$)	Highly selective
Set-C Drug-Likeness (n = 17)		
Lipinski compliance	100 %	Excellent
Veber compliance	100 %	Excellent
Mean QED	0.703	Good
Mean logP	2.84	Favorable
Mean TPSA	66.3 Å^2	Good permeability

Table S5: RRS class definitions used in the integrated analysis.

Class	Operational definition	Number
A*	maximum absolute eligible WT score $\geq 7\,\mathrm{kcal\,mol^{-1}}$ and all available RRS values $\geq 80\,\%$	7
A	All available RRS values $\geq 80\,\%$, without the A* wild-type criterion	0
B	All available RRS values $\geq 70\,\%$, but not all $\geq 80\,\%$	4
C	At least one RRS value $\geq 80\,\%$, with the complete-panel criteria for A/A* or B not met	6
D	No available RRS value $\geq 80\,\%$	0

two-sided empirical $p = 0.0019$, consistent with the analytical result.

A sensitivity analysis controlling for molecular weight and Murcko-scaffold prevalence

Table S6: Wild-type and mutant docking scores (kcal mol^{-1}) underlying the RRS values of Table 1. PfDHFR values come from the separate mutation-docking panel (7F3Y; exhaustiveness 128); PfCRT values come from the corrected-protocol re-dock against the 3D7-like LYS-76 wild-type and the K76T/K76A mutants on the cavity-anchored V2 grid (exhaustiveness 32; Section S12.7). RRS is the percentage ratio of the absolute mutant to the absolute wild-type score (Section S7); wild-type non-binders ($|S_{\mathrm{Vina},i,WT,t}| < 5$ kcal mol^{-1}) are shaded by dashes in the RRS table and make no contribution to RRS$_{\mathrm{mean}}$. Scores are AutoDock Vina scoring-function outputs, not experimental free energies.

Candidate	PfDHFR (kcal mol^{-1})					PfCRT (kcal mol^{-1})		
	WT	N51I	C59R	S108N	I164L	WT	K76T	K76A
PP-01	−7.50	−6.48	−6.41	−6.57	−6.29	−9.15	−9.14	−9.15
PP-02	−1.10	−6.52	−0.88	−6.41	−6.63	−9.34	−9.36	−9.34
PP-03	−10.30	−7.01	−7.04	−6.78	−7.11	−12.01	−12.01	−12.01
PP-04	−8.00	−5.50	−5.35	−5.50	−5.88	−7.40	−7.39	−7.42
PP-05	−0.70	−6.36	−6.26	−6.33	−6.29	−8.96	−8.96	−8.97
PP-06	−0.60	−6.71	−7.45	−6.76	−6.88	−9.03	−9.03	−9.03
PP-07	−7.60	−5.46	−5.46	−5.50	−5.49	−7.95	−7.93	−7.95
PP-08	−7.60	−4.72	−4.72	−5.11	−5.04	−6.59	−6.57	−6.60
PP-09	−7.10	−5.01	−5.18	−5.38	−5.43	−7.44	−7.42	−7.44
PP-10	−7.70	−5.85	−5.88	−5.83	−5.84	−8.04	−8.05	−8.06
PP-11	−0.70	−6.76	−0.02	−6.89	−6.82	−9.34	−9.35	−9.34
PP-12	−7.50	−5.64	−4.96	−4.72	−5.44	−8.06	−8.06	−8.06
PP-13	−3.40	−7.32	−6.56	−7.10	−6.14	−9.79	−9.70	−9.73
PP-14	−7.60	−4.73	−4.98	−5.17	−4.93	−6.43	−6.43	−6.43
PP-15	−5.60	−6.90	−6.31	−7.05	−7.38	−9.81	−9.81	−9.82
PP-16	−7.60	−5.54	−5.68	−5.60	−5.77	−7.47	−7.51	−7.44
PP-17	−8.30	−4.92	−5.08	−4.94	−4.89	−6.37	−6.38	−6.38

shows that the PNS–RRS association persists (partial $\rho = -0.6154$ versus raw $\rho = -0.2098$, $n = 12$), so the negative direction reported here is not a size or scaffold artifact. The N_{fav}–RRS association survives Bonferroni correction for the three primary tests ($\alpha = 0.017$): candidates with more favorable targets tend to carry higher mean RRS.

At $n = 17$, a two-sided test at $\alpha = 0.05$ has 80 % power for $|\rho| \geq 0.62$ (asymptotic approximation). The observed $\rho = 0.714$ clears this threshold; weaker associations remain undetectable at this sample size.

Table S7: Retrospective RRS profiles of five approved antimalarials against the PfDHFR (MTX-stripped wild-type and N51I/C59R/S108N/I164L mutants) and corrected PfCRT (LYS-76 wild-type and K76T/K76A mutants) panels. RRS is defined as in Section S7: $\mathrm{RRS}_{i,m,t} = 100|S_{\mathrm{Vina},i,m,t}|/|S_{\mathrm{Vina},i,WT,t}|$; blank cells mark wild-type non-binders ($|S_{\mathrm{Vina},i,WT,t}| < 5\,\mathrm{kcal\,mol^{-1}}$) for which no RRS is defined. The docking-RRS framework does not recover the clinically documented resistance signatures: pyrimethamine RRS is $\approx 100\,\%$ on all four PfDHFR mutants and chloroquine RRS is $\approx 99\,\%$ on PfCRT K76T/K76A, both within the pipeline-null spread of $99.4\,\%$ to $100.5\,\%$ (Section S12.7). Artemisinin and cipargamin show PfDHFR RRS values above $100\,\%$ (mutant estimates more favorable than wild-type); chloroquine and lumefantrine are PfDHFR non-binders, consistent with their non-antifolate mechanisms. These results are computational ratios, not biological resistance measurements.

Drug	$\mathrm{RRS}_{\mathrm{PfDHFR}}$ (%)				$\mathrm{RRS}_{\mathrm{PfCRT}}$ (%)	
	N51I	C59R	S108N	I164L	K76T	K76A
Artemisinin	113.5	118.0	113.8	114.0	100.0	100.0
Pyrimethamine	100.0	100.0	99.7	100.0	100.2	100.1
Chloroquine	—	—	—	—	99.6	98.8
Lumefantrine	—	—	—	—	105.8	103.5
Cipargamin	115.1	112.7	114.6	114.6	99.5	99.7

Table S8: Per-target Vina score distributions for the 17-member cohort on the corrected matrix (PfCRT re-docked against the corrected 3D7-like LYS-76 receptor on the cavity-anchored V2 grid). Favorability is defined within each target as a Vina estimate at or below that target's median (per-target medians: PfDHFR −5.92, PfCRT −8.06, PfClpP −6.02, PfATP4 −6.38 kcal mol^{-1}), avoiding a single cross-target cutoff. The threshold is a prioritization device, not an affinity cutoff.

Target	Min	Median	Mean	Max	n
PfDHFR	−6.346	−5.921	−5.755	−4.863	17
PfCRT	−12.010	−8.056	−8.422	−6.374	17
PfClpP	−7.031	−6.016	−5.935	−5.047	17
PfATP4	−7.565	−6.378	−6.308	−4.635	17
All targets	**-12.010**	**-6.316**	**-6.605**	**-4.635**	**68**

S9 Target evidence classes and limitations

The panel spans four distinct antimalarial vulnerabilities (anchors and evidence classes in Section 2.3): PfDHFR represents folate metabolism and antifolate resistance; PfCRT represents a resistance-linked digestive-vacuole transporter; PfClpP represents parasite proteostasis through the caseinolytic protease, a vulnerability supported by recent work on

Table S9: Spearman correlations on the 17-member cohort. Bonferroni-adjusted threshold: $\alpha = 0.017$ for the three hypothesis-oriented comparisons.

Pair	ρ	p	Interpretation
PNS–RRS	−0.714	0.001 3	Bonferroni-significant ($p < 0.017$)
ACSI–RRS	−0.190	0.465	Not significant
RRS–wild-type Vina score	0.691	0.002 1	Bonferroni-significant ($p < 0.017$)
PNS–wild-type Vina score	−0.375	0.138	Not significant
N_{fav}–$\mathrm{RRS}_{\mathrm{mean}}$	0.714	0.001 3	Bonferroni-significant ($p < 0.017$)

Table S10: Sensitivity of the PNS ranking to the imputed PfCRT centrality. The reference value is the observed network-mean imputation ($C_{\mathrm{PfCRT}} = 0.151$). Alternative values were zero, one-half, one-and-a-half, and twice the reference. Spearman ρ is calculated against the reference 17-candidate ranking; all candidates have two target entries in the PNS record. The ranking is sensitive to this network-data choice; target essentiality and biological polypharmacology were not assessed.

Imputation	C_{PfCRT}	Spearman ρ	Minimum PNS	Maximum PNS	n
Zero	0.000 0	0.971 6	0.300	5.147	17
One-half canonical	0.075 5	0.997 5	0.674	5.573	17
Canonical network mean	0.151 0	1.000 0	1.047	6.000	17
One-and-a-half canonical	0.226 5	0.975 5	1.421	6.426	17
Twice canonical	0.302 0	0.963 2	1.795	6.853	17

apicoplast chaperonin–Clp proteostasis;[S3] and PfATP4 represents P-type ATPase-mediated ion regulation, for which an endogenous structure and modulator-bound state have recently been reported.[S4] PfDHFR uses a catalytic-site ligand anchor; PfClpP uses a catalytic triad; PfCRT uses a membrane-associated Y01 cavity proxy; and PfATP4 uses conserved P-type ATPase machinery without a co-crystallized inhibitor. Consequently, all four-target profiles are exploratory and are not equivalent evidence of biological target engagement. No RRS values are reported for PfClpP or PfATP4.

S10 Multi-parameter optimization framework for upstream candidate selection

The 17-member Set-C cohort analyzed in this study was selected from a larger upstream library using a multi-parameter optimization (MPO) framework that integrated docking scores, drug-likeness, and ADMET predictions. The MPO components are formulated below (concise summary in the main text, Section 2.1); the full MPO validation (sensitivity analysis, centroid-based screening workflow, and benchmark enrichment) is described in Section 2.1 and Section S10. Set-C candidates came from the MPO $\geq$ 0.40 tier; MPO scores are computational prioritization, not confirmed biological activity or synthetic feasibility.

S10.1 MPO scoring components

The total MPO score for compound i is defined as:

$$\mathrm{MPO}_{\mathrm{total},i} = \sum_{k=1}^{5} w_k \cdot S_{k,i} + \text{Polypharmacology Bonus}_i \tag{1}$$

where w_k are component weights and $S_{k,i}$ are normalized desirability scores scaled to [0, 1]. The five components and their weights are:

Component 1: Vina binding affinity (35 % weight). Min-max scaled:

$$S_{\mathrm{vina},i} = \frac{\mathrm{Vina}_i - \mathrm{Vina}_{\min}}{\mathrm{Vina}_{\max} - \mathrm{Vina}_{\min}} \tag{2}$$

where more negative values yield higher desirability.

Component 2: DiffDock confidence (25 % weight). Logistic sigmoid transformation:

$$S_{\mathrm{diff},i} = \frac{1}{1 + e^{-c_i}} \tag{3}$$

where c_i is the DiffDock confidence score.[S5] High-confidence poses ($c > 0$) approach 1.0;

low-confidence poses ($c < -1.5$) approach 0.

Component 3: Quantitative estimate of drug-likeness (QED) (20 % weight). Direct use:

$$S_{\mathrm{qed},i} = \mathrm{QED}_i \tag{4}$$

Component 4: ADMET composite (15 % weight). Average of 11 desirability endpoints:

$$S_{\mathrm{admet},i} = \frac{1}{11}\sum_{m=1}^{11} d_m(\mathrm{ADMET}_{i,m}) \tag{5}$$

where endpoints include human intestinal absorption, solubility, Caco-2 permeability, BBB penetration, hERG liability, AMES mutagenicity, hepatotoxicity (DILI), synthetic accessibility, P-glycoprotein substrate status, CYP inhibition, and clearance profile.[S6]

Component 5: Lipinski penalty (5 % weight). Proportional to violations:

$$P_{\mathrm{Ro5},i} = 1 - 0.2 \cdot n_{\mathrm{violations},i} \tag{6}$$

where $n_{\mathrm{violations},i} \in [0, 4]$ (MW $> 500\,\mathrm{Da}$, $\log P > 5$, HBD > 5, HBA > 10).

Polypharmacology bonus. Compounds binding multiple targets receive:

$$\text{Polypharmacology Bonus}_i = 0.05 \cdot (n_{\mathrm{targets},i} - 1) \quad \text{if } n_{\mathrm{targets},i} \geq 2 \tag{7}$$

capped at 0.10 (3 targets). The $\mathrm{MPO}_{\mathrm{target}} \geq 0.50$ threshold ensures only candidates meeting minimum quality criteria qualify for the bonus.

S10.2 Weight rationale and scope

The 35 % Vina and 25 % DiffDock weights (combined 60 %) prioritize target-engagement evidence. 20 % QED ensures drug-likeness, 15 % ADMET addresses early toxicity, and the 5 % Ro5 penalty reflects Lipinski compliance already partly captured within QED. Component weights were selected based on established drug-discovery priorities[S7–S9] rather than target-specific optimization. The MPO framework served as an upstream filter to select Set-C candidates from the 65 856-molecule library; it does not validate Set-C quality.

The complete MPO methodology, including centroid-based screening workflow (484 centroids → 76 optimization-worthy → 53 clusters → 19 913 synthesizable leads), sensitivity analysis (25 weight perturbations, Spearman $\rho = 0.792$ for top-1000), and external validation (DEKOIS, MMV Malaria Box enrichment), is described in Section 2.1 and Section S10.

S11 Reproducibility and availability

The candidate manifest, docking and RRS tables, cross-metric matrix, scripts, receptor/ligand files, and run records are archived on Zenodo under an CC BY 4.0 license at https://doi.org/10.5281/zenodo.22686176. The figures and derived tables are reproducible computational outputs; they do not substitute for independent structural or experimental validation.

S12 Docking validation and protocol assessment

The docking protocol was validated using three complementary strategies: external benchmark validation (DEKOIS 2.0), positive-control enrichment (MMV Malaria Box), and pose reproduction accuracy (redocking of co-crystallized ligands; Section 2.3.1). Together they expose both the capabilities and the limits of the protocol.

S12.1 PfCRT receptor correction

The deposited 6UKJ cryo-EM structure (7G8 isoform) carries threonine at residue 76 of chain A, corresponding to the K76T chloroquine-resistance substitution. Because the study requires a wild-type PfCRT reference for the RRS analysis, residue 76 was corrected to lysine (3D7-like wild-type) before docking. The corrected receptor was validated by P2Rank pocket prediction:[S10] the top-ranked pocket (score 162.7, probability 0.999) lies 5.7 Å from the Vina grid center, confirming that the grid box covers the binding site. The corrected LYS-76 receptor was used for the PfCRT re-docking (Section S12.7) on the V2 grid (center $(152.99, 151.042, 159.379)$, 25 Å box) used for PfCRT throughout.

S12.2 External benchmark and enrichment

Table S11: Docking validation: external DEKOIS benchmark and MMV Malaria Box enrichment. *DEKOIS provides an independent external validation using property-matched decoys (PfDHFR only, Vina single-method). The near-random DEKOIS ROC-AUC (0.450) establishes the baseline limitation of docking-only scoring. The two controlled arms (exhaustiveness 32, identical grid and ligands; receptor with MTX retained vs MTX-stripped) show that removing the retained MTX does not restore enrichment: bootstrap confidence intervals overlap and EF@1% is zero in both arms. MMV Malaria Box values use consensus Vina+DiffDock scoring; for PfCRT and PfATP4, active/decoy labels derive from score strata and therefore measure retrodictive consistency rather than external discrimination.*

Target (PDB)	ROC-AUC	EF@1%	EF@5%	EF@10%	BEDROC	PR-AUC	Actives	Decoys
DEKOIS 2.0 external validation (Vina only):								
PfDHFR (7F3Y)	0.450	0.00	0.00	1.00	0.021	0.034	40	1 200
DEKOIS 2.0 controlled comparison, exhaustiveness 32 (Vina only):								
PfDHFR (7F3Y, MTX retained)	0.502	0.00	0.50	0.25	0.026	0.032	40	1 199
PfDHFR (7F3Y, MTX-stripped)	0.563	0.00	1.00	1.00	0.048	0.038	40	1 199
MMV Malaria Box positive-control (Vina+DiffDock consensus):								
PfDHFR (7F3Y)	0.924	2.70	2.57	2.72	1.309	0.842	399	399
PfCRT (6UKJ)	0.971	1.11	1.11	1.11	9.434	0.992	359	40
PfATP4 (9N10)	1.000	2.05	2.00	2.00	1.810	1.000	99	99

Notes: Top: DEKOIS 2.0 PfDHFR benchmark with 40 experimentally validated actives and 1 200 property-matched decoys, docked with AutoDock Vina (single-method, no consensus). The near-random ROC-AUC (0.450, 95 % CI [0.367 to 0.531]) establishes an external baseline for docking-only limitations and motivates the consensus strategy. Middle: controlled two-arm comparison (2026-09-09) on the deposited 7F3Y receptor with MTX retained versus an MTX-stripped receptor in the same coordinate frame, same grid and ligand preparation, exhaustiveness 32; bootstrap ROC-AUC 95 % CIs are [0.423 to 0.586] (retained) and [0.477 to 0.646] (stripped); one decoy failed ligand preparation identically in both arms. Bottom: MMV Malaria Box enrichment using Vina+DiffDock consensus scores; PfCRT and PfATP4 labels are score-stratified (top/bottom 25 %), measuring retrodictive consistency rather than external discrimination. BEDROC calculated with $\alpha = 20$. EF = Enrichment Factor, PR-AUC = Precision-Recall Area Under Curve.

The DEKOIS 2.0 benchmark[S11] provides property-matched decoys that avoid trivial separation in naïve active/decoy sets. For PfDHFR (7F3Y), AutoDock Vina alone gives ROC-AUC 0.450 [95 % CI: 0.367 to 0.531]: near-random, reflecting the known limits of single-method scoring. To test whether the MTX retained in deposited 7F3Y blocks the antifolate pocket and explains this null enrichment, we ran the identical DEKOIS panel as a controlled two-arm comparison (same 40 actives, 1 200 decoys, and grid) against two receptors: the deposited 7F3Y receptor with MTX retained (ROC-AUC 0.502 [95 % CI: 0.423 to 0.586], EF@1% 0.00, EF@5% 0.50) and an MTX-stripped receptor rebuilt in the same coordinate frame (ROC-AUC 0.563 [95 % CI: 0.477 to 0.646], EF@1% 0.00, EF@5% 1.00), at exhaustiveness 32 in both arms (the revised-protocol standard). The confidence intervals overlap and EF@1% stays zero in both arms: removing MTX gives a marginal, non-significant gain (ΔAUC = 0.060) without restoring enrichment, so retained-MTX blockade is not supported. The retained arm matches the original exhaustiveness-64 baseline, indicating that the near-chance result is robust to protocol details (Table S11).

Hany and co-workers applied the same DEKOIS 2.0 protocol to wild-type and quadruple-mutant PfDHFR with AutoDock Vina and showed that machine-learning rescoring (CNN-Score, RF-Score-VS) lifted performance from worse-than-random to better-than-random:[S12] the closest prior assessment of this baseline, and independent support for rescoring over raw Vina ranking.

The MTX redocking failure (RMSD $\approx$ 30 Å), together with the controlled comparison above, points to grid position (near the NADPH cofactor site) and, predominantly, single-method scoring limits rather than steric blockade by the retained ligand. The primary library screening therefore used a consensus Vina+DiffDock strategy, which enriched substantially better (ROC-AUC 0.924 on the MMV Malaria Box positive-control set). The PfCRT and PfATP4 MMV enrichments use score-stratified labels (top/bottom 25 % of the docked library), so they measure retrodictive consistency rather than external discrimination.

S12.3 Redocking validation

Co-crystallized ligands were reproduced (RMSD <2.0 Å) in 4 of 5 cases (80 %). The single failure occurred for methotrexate (MTX) at PfDHFR, where the grid center was positioned near the NADPH cofactor binding site rather than the folate substrate pocket, showing target-specific grid-position sensitivity, which motivates anchor-guided box placement. Full-

Table S12: Redocking validation: reproduction of co-crystallized ligand poses. *Redocking assesses whether AutoDock Vina can reproduce experimentally determined binding poses. RMSD <2.0 Å indicates successful pose reproduction. Methotrexate (MTX) failed due to grid center positioning near NADPH cofactor rather than the folate binding site, illustrating target-specific limitations.*

Target	PDB	Ligand	RMSD (Å)	Result
PfDHFR	7F3Y	MTX (methotrexate)	$\sim$30 Å	Failed (grid centered on NADPH)
PfDHFR	7F3Y	P218 (pyrimethamine)	1.42	Success
PfCRT	6UKJ	Y01 (proxy ligand)	1.78	Success
PfClpP	2F6I	Peptide fragment	1.65	Success
PfATP4	9N10	ADP (cofactor)	1.23	Success
Success rate: 4/5 alignable ligands (80 %)				

Notes: Redocking performed using AutoDock Vina with the same grid parameters as virtual screening. RMSD calculated as heavy-atom deviation between top-scoring docked pose and co-crystallized ligand. MTX redocking failed because the grid was centered on the NADPH cofactor binding site ($\sim$12 Å from folate pocket), not the MTX binding site; this illustrates target-specific validation requirements. PfCRT Y01 is a membrane-cavity proxy ligand (not a validated inhibitor). Success criterion: RMSD <2.0 Å.

ligand redocking succeeded for PfDHFR (P218) and PfCRT (Y01); PfClpP (2F6I) was validated by peptide-fragment pose reproduction only, and PfATP4 (9N10) by ADP cofactor pose reproduction only. These partial validations are not equivalent to full-ligand redocking.

S12.4 Multi-seed validation

Mode-1 affinity estimates are stable to seed choice within each protocol (maximum spread 0.054 kcal mol^{-1} across the four five-seed rows; Table S13). The PP-15 PfCRT runs on the corrected LYS-76 receptor give scores (-7.81 to -7.84), broadly consistent with the single-run matrix value (-9.81 kcal mol^{-1} on the cavity-anchored V2 grid, Table S1) given the protocol differences; the PP-01 canonical PfCRT score falls within its 5-seed spread. These runs used the revised canonical-grid protocol and therefore report seed dispersion for that protocol; differences from the single-run matrix are listed in the table (Section S3). The matrix in Table S1 underlies the main-text PfCRT RRS and favorability calculations and was additionally checked with the pipeline-null control (Section S12.7).

Table S13: Multi-seed docking sensitivity for PP-15 and PP-01 against PfDHFR and PfCRT wild-type receptors. Five independent AutoDock Vina runs with different random seeds (exhaustiveness 32 for PP-15, 128 for PP-01 canonical) assess seed sensitivity under the revised canonical-grid protocol. These runs use grid centers and receptor preparations that differ from the single-run matrix in Table S1 (e.g., PP-15 PfDHFR −8.59 here versus −6.05 in Table S1; see footnote), so they report seed dispersion for their own protocol, not for the Table S1 single-run scores.

Candidate	Target	Receptor	Seeds (5 runs)	Mean ± SD	Range	Seed dispersion
PP-15	PfDHFR WT	7F3Y	−8.585, −8.588, −8.603, −8.589 and −8.586	−8.590(7)	0.018	<0.02 k
PP-15	PfCRT WT	PfCRT_WT.pdb (LYS-76)	−7.840, −7.823, −7.810, −7.820 and −7.816	−7.822(11)	0.030	<0.03 k
PP-01	PfDHFR WT	7F3Y	−7.529, −7.504, −7.485, −7.504 and −7.475	−7.499(21)	0.054	<0.06 k
PP-01	PfCRT WT	6UKJ (deposited, THR-76)	−9.257, −9.262, −9.257, −9.225 and −9.250	−9.250(15)	0.037	<0.04 k

Protocol disclosure: These five-seed runs were executed under the revised canonical-grid protocol (PfDHFR grid centered at $(1.33, -1.733, -23.842)$; PP-15 PfCRT grid centered at $(149.821, 169.427, 141.111)$; PP-01 PfCRT grid centered at $(152.99, 151.042, 159.379)$; all 25 Å boxes). They are not directly comparable to the single-run matrix in Table S1, which was generated under the original panel grids; the difference between the two protocols is visible in the PP-15 PfDHFR scores (−8.59 here versus −6.05 in Table S1). The PP-15 PfCRT runs used the corrected LYS-76 receptor; the PP-01 PfCRT runs used the deposited 6UKJ receptor. Seed dispersion (<0.06 kcal mol^{-1} across all four rows) shows that mode-1 affinity estimates are stable to seed choice within each protocol. Seed-stability assessment of the revised Table S1 matrix under the corrected-receptor protocol is provided together with the corrected-receptor re-docking results in the response letter.

S12.5 Validation dataset summary

S12.6 Validation interpretation and scope

Together, DEKOIS, MMV enrichment, and redocking show that the consensus Vina+DiffDock protocol gives reliable pose sampling and enrichment for ranking, within the limits of physics-based docking. The near-random single-method DEKOIS baseline confirms that scoring-function limitations persist and that enrichment depends on consensus strategies. Consequently, reported Vina scores support ranking, not experimental-affinity claims.

S12.7 PfCRT re-docking and pipeline-null control

The PfCRT channel was re-docked on a common coordinate frame with two wild-type-equivalent receptors and two mutant receptors: (i) the corrected 3D7-like wild-type (LYS-76, rebuilt from the deposited 6UKJ chain A by residue-76 correction); (ii) a pipeline-processed

Table S14: Summary of validation datasets used for docking protocol assessment. *Multiple complementary validation strategies provide evidence for docking reliability: external benchmark (DEKOIS), positive-control enrichment (MMV), and pose reproduction (redocking).*

Target	PDB	Actives	Decoys/Negatives	Validation Type
External benchmark validation:				
PfDHFR	7F3Y	40	1 200	DEKOIS 2.0 (property-matched)
Positive-control enrichment:				
PfDHFR	7F3Y	399	399	MMV Malaria Box (consensus)
PfCRT	6UKJ	359	40	MMV (score-stratified)
PfATP4	9N10	99	99	MMV (score-stratified)
Redocking validation:				
PfDHFR	7F3Y	2 ligands, 1 success (MTX failed: grid on NADPH)		Full-ligand redocking
PfCRT	6UKJ	1 ligand (1 success)		Full-ligand redocking
PfClpP	2F6I	1 ligand, peptide fragment (RMSD 1.65 Å)		Fragment pose reproduction
PfATP4	9N10	1 ligand, ADP cofactor (RMSD 1.23 Å)		Cofactor pose reproduction

Notes: DEKOIS 2.0 provides independent external validation with carefully designed property-matched decoys, addressing trivial separation artifacts. MMV Malaria Box positive-control uses experimentally confirmed antimalarials; PfCRT and PfATP4 labels are score-derived (top/bottom 25 %) and measure retrodictive consistency. Redocking assesses pose reproduction accuracy (RMSD <2.0 Å). PfClpP (2F6I) and PfATP4 (9N10) have no co-crystallized small-molecule inhibitor; their "1 success" reflects fragment/cofactor pose reproduction, not full-ligand redocking, and should not be interpreted as equivalent to the full-ligand successes for PfDHFR and PfCRT. The MTX redocking failure (RMSD ≈30 Å) may partly reflect grid-positioning bias (NADPH-adjacent site) rather than scoring-function limitations alone. These complementary checks support protocol reliability within the stated limits (DEKOIS ROC-AUC 0.450 baseline).

wild-type control (LYS-76 processed identically but unmutated); (iii) the K76T mutant; and (iv) the K76A mutant. All four receptors were prepared with the same mutation and protonation pipeline (`mutate_pdb.py` plus Open Babel gasteiger charges) so that the only difference between the null and the mutants is the intended substitution. All 17 candidates were docked against each receptor on the cavity-anchored V2 grid (center $(152.99, 151.042, 159.379)$, 25 Å box, exhaustiveness 32), giving 68 runs (17 ligands $\times$ 4 receptors). The corrected wild-type scores form the PfCRT column of the four-target matrix and feed the PfCRT RRS values and per-target-median favorability counts.

The RRS ratios were recomputed as RRS $= 100|S_{\text{Vina},i,m}|/|S_{\text{Vina},i,WT}|$ against the corrected LYS-76 wild-type denominator. K76T and K76A ratios span 99.1 % to 100.6 % and 99.4 % to 100.4 % respectively, and the pipeline-null ratios span 99.4 % to 100.5 %. The null distribution therefore brackets the 80 %/70 % class boundaries entirely: the maximum deviation of any mutant ratio from the null ratio is 1.2 percentage points (K76T) and 0.5 percentage points (K76A), both far below the 10 % boundary spacing. The corrected PfCRT channel consequently carries no detectable K76T/K76A effect for this panel, and PfCRT RRS values near 100 % are read as neutral-within-null rather than as evidence of active resistance circumvention (Table 1; Section 3.4).

The neutral-within-null reading is consistent with an extended replication panel (39 ligands $\times$ 8 PfDHFR/PfCRT states, 312/312 finite records, prepared independently for this study), whose bootstrap RRS distribution of 100.45 [99.59, 101.32] brackets the corrected channel's null spread; in a score-perturbation sensitivity analysis, class assignments are stable to bounded $\pm 1\,\text{kcal}\,\text{mol}^{-1}$ (unchanged in 212/272 records, 77.9%).

S13 Available machine-readable resources

The Zenodo deposit contains:

- the candidate manifest and canonical molecular identifiers;
- the complete target-wise Vina score matrix, ligand poses, and run records;
- the wild-type/mutant docking panel used for the per-target RRS analysis;
- the derived RRS, PNS, ACSI, and cross-metric tables;
- the chemical-space similarity data and source data for the coverage figure;
- receptor and ligand input files, configuration files, analysis scripts, and figure-generation scripts;

- checksums and reproducibility metadata.

Exact file names, checksums, and version-specific dependencies are in the deposit index. The main-text evidence boundary is in Section 2.6.

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